AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI

ANNUAL EXAMINATIONS (THEORY) 2024

Computer Science Paper II

Time: 1 hours 30 minutes Marks: 35

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INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's Signature**

RUBRIC

2. There are EIGHT questions. Answer ALL questions. Question 7 and 8 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid, or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a scientific calculator if you wish.

Q.1. (Total 6 Marks)

Your teacher asked you to type a list of your favourite animals using a computer. First, you type the names of the animals using the keyboard, then the computer sorts the names in alphabetical order, and show you the list on the monitor.

i. Name the stages of data processing cycle that were involved in the given scenario. (3 marks)

ii. Describe each stage that you have named. (3 marks)

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Q.2. (Total 3 Marks)

If we talk about computer memory, lower access time is usually better.

Give THREE reasons to justify this statement.

_____

_____

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Q.3. (Total 3 Marks)

Differentiate Intel Pentium IV processors from Advance Micro Devices (AMD) Athlon processors based on the given characteristics.

	Intel Pentium IV Processor	**AMD Athlon Processor**
Clock Speed		
Cache		
Architecture		

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Q.4. (Total 3 Marks)

Write any THREE primary functions of High Definition Multimedia Interface (HDMI) port that makes it different from PS/ 2 port.

Q.5. (Total 3 Marks)

i. Write the full form of Wi-Fi. (1 mark)

ii. Which wireless medium allows Wi-Fi devices to connect to the internet or each other without using wires? (1 mark)

iii. Name any ONE place where Wi-Fi is commonly used. (1 mark)

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Q.6. (Total 3 Marks)

Compare Database Management System (DBMS) and traditional File Management System based on the given aspects.

Aspect	Database Management System (DBMS)	Traditional File Management System
Data Security		
Data Retrieval		
Data Redundancy		

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Q.7. (Total 7 Marks)

EITHER

a.

i. Define unguided transmission. (1 Mark)

ii. Write TWO advantages and TWO disadvantages of using unguided media. (4 Marks)

iii. Name and describe ONE type of unguided media used for wireless communication. (2 Marks)

OR

b.

i. Describe Geostationary Earth Orbit (GEO). (2 Marka)

ii. Write TWO advantages of GEO satellite. (2 Marks)

iii. Give THREE applications of GEO satellite. (3 Marks)

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Q.8. (Total 7 Marks)

EITHER

a. Consider the given image showing three tables in an MS Access database of a library.



Books can have multiple authors, and users can borrow multiple books.

Draw an entity-relationship diagram (ERD) for this database. (7 Marks)

Space for ERD

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OR

Consider the given table named as Product.

Product_ID	Product_Name	Price	Stock_Quantity	Category
P101	Penu Chicken Nuggets	1580.00	74	Frozen Item
P102	Tesle Milkrak	1920.50	25	Dairy
P103	Mapple Smart Watch X	19500.90	54	Electronics

b. Write MS Access SQL queries to

i. create this table with appropriate data types and with the following field lengths. Set Product_ID as primary key. (4 Marks)

Field Name	Field_Length
Product_ID	5
Product_Name	25
Category	20

ii. remove the "Category" field from the "Product" table. (1 Mark)

iii. add a new field called "Brand" to the "Product" table with a field length of 15. (1 Mark)

iv. change the field length of "Product_Name" so that it should accept up to 40 characters. (1 Mark)

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END OF PAPER

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